'98

銀河天文学特論II (Oct 5 ~)

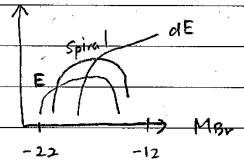
M32 compact eliptical

イントロ

NO. 1

D Oct 19

dE dwarf eliptical



Field Galaxy

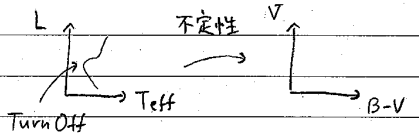
early type

late type

Vergo cluster

1.2 銀河の年齢

(1) 球状星団の年齢



$m_V \rightarrow M_V$ (Turn off) $\rightarrow$ age

distance
ISM reddening
} theory

$\Delta V^{\text{Turn off}}_{\text{Horizontal Branch}}$ (t_{age}, [Fe/H])

model $\rightarrow M_V$(TO) (t_{age}, Fe/H)

M_V(HB)
M_V(RR)

$\downarrow$

$\Delta V^{TO}_{HB} \leftrightarrow$ obs ΔV^{TO}_{HB}

(t, Fe/H) $\rightarrow$ age

M_V(RR) (or M_V(HB))

① Subdwarf MS fitting ← distance ; known

太陽近傍の subdwarfs の Color-Magnitude diagram

→ subdwarf の MS が決まる

→ 同じ [Fe/H] をもつ cluster の MS の決定

→ M_V(RR) (距離)

② LMC の RR Lyrae

$(m-M)_0 = 18.5$ mag
V(RR)
} → M_V(RR)

③ field RR Lyrae stars.

(太陽近傍) → M_V(RR)

distance は ヒッパルコス

→ $t_{age} \simeq 11.5$ Gyr

④ WD sequence fitting

(2) ハロー星の年齢 $t_{age} \approx 18$ Gyr

(3) high-z 銀河の年齢

1.3 膨張宇宙の重力学

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NO. 2

2. 膨張宇宙における構造形成

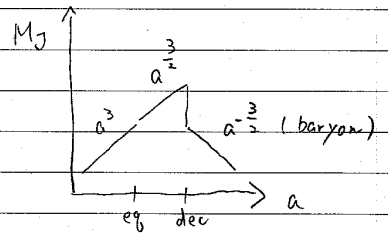
Oct 26

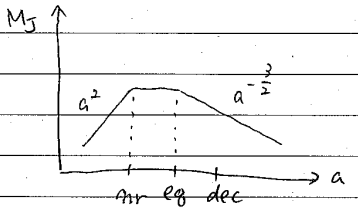
2.1 重力不安定性の線形理論

2.1.1 fluid dynamics

$$\nabla_r P = \frac{d\rho}{dr}\frac{dP}{d\rho} = c_s^2 \frac{d\rho}{dr}$$

Nov 2





$\begin{cases} \rho = \rho_{DM} \\ P = P_{rad} \leftarrow \text{DMの pressure} \end{cases}$

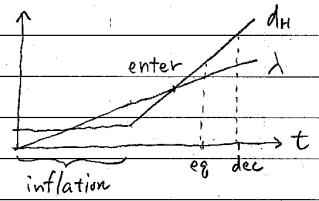
δ_{DM} の成長の仕方　　$\lambda \propto a \propto t^n$　　$n<1$

$d_H \propto t$

HDM

free streaming

silk mass だったっけ？

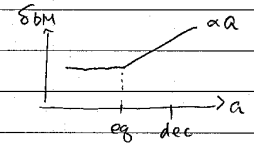


$t_{exp} \propto 1/\sqrt{G\rho_{rad}} \propto a^2$ ← $t_{exp} > t_{collapse}$

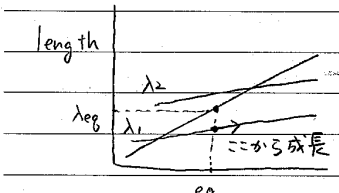
$t_{collapse} \propto 1/\sqrt{G\rho_{DM}} \propto a^{3/2}$

$\rightarrow \delta_{DM} \propto a$

$\delta_{DM} \propto \ln a$



2.1.3 Power spectrum of DM fluctuations



(1) $t_{enter} < t_{eq}$

$\lambda < \lambda_{eq}$

$$\delta_{DM} \sim \begin{cases} \text{const} = \delta_{DM}(t_{enter}) & a < a_{eq} \\ \delta_{DM}(t_{enter})\left(\frac{a}{a_{eq}}\right) \propto a & a_{eq} < a \end{cases}$$

(2) $t_{enter} > t_{eq}$

$\lambda > \lambda_{eq}$

$M_{eq} \sim 10^{14} M_\odot$

$$\delta_{DM} = \delta_{DM}(t_{enter})\left(\frac{a}{a_{enter}}\right) = \delta_{DM}(t_{enter})\left(\frac{M_{eq}}{M}\right)^{2/3}\left(\frac{a}{a_{eq}}\right)$$

$|\delta|^2 = |\delta_k|^2 d^3k = 4\pi k^2 dk$

$= k^3|\delta_k|^2 = M^{-1}|\delta(\lambda)|^2$

$= \begin{cases} M^{\frac{2}{3}\alpha-1}\left(a/a_{eq}\right)^2 & M < M_{eq} \\ M^{\frac{2}{3}\alpha-1-\frac{4}{3}}\left(a/a_{eq}\right)^2 & M > M_{eq} \end{cases}$

$k^3|\delta_{DM}(t_{enter})|^2$

$\delta_\lambda(t_{enter}) \propto \lambda^\alpha \propto k^{-\alpha} \propto M^{\alpha/3}$

$\delta_\lambda \propto \begin{cases} M^{\alpha/3}\left(\frac{a}{a_{eq}}\right) & \lambda < \lambda_{eq} \\ M^{\frac{\alpha-2}{3}}\left(\frac{a}{a_{eq}}\right) & \lambda > \lambda_{eq} \end{cases}$

$\alpha = \frac{3}{2}$: M に indep (scale inv)

$\alpha = \frac{3}{2}k^3|\delta_k|^2 \sim \begin{cases} \text{const}\left(a/a_{eq}\right)^2 & M < M_{eq} \\ M^{-4/3}\left(a/a_{eq}\right)^2 & M > M_{eq} \end{cases}$

$\left(\ln \frac{M_{eq}}{M}\right)^2$

$|\delta_R(t_i)|^2 \propto k^n$　　$n=1$　HZ

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$$P(k,t_f) = \left[\frac{b(t_i)}{b(t_f)}\right]^2 T^2(k,t_f)\, P(k,t_i)$$

$b(t_f)$: linear growth

$P(k,t_i) = |\delta k|^2 = Ak$

Transfer function

$$T_{CDM}(k) = \frac{\ln(1+2.34q)}{2.34q}\left[1+\cdots\right.$$

$q = k/\Omega_0 h^2$ Mpc

(Bardeen et al 1986 ApJ. 304, 15-)

$$T_{HDM}(k) = 10^{-(k/k_{FS})^{1.5}} \qquad k_{FS} = 0.16\left(\frac{m_\nu}{30eV}\right) \text{Mpc}^{-1} \qquad M_{FS} = 4\times10^{15}\left(\frac{m_\nu}{30eV}\right)^{-2} M_\odot$$

- MDM = CHDM = CDM + HDM (Mixed) 2/3 1/3 $\mu \to \tau$ $\Omega_\nu = m/92\, h^2$ eV, 5eV
- $n \lesssim 1$ + CDM
  tilted CDM

2.2 非線形段階における密度ゆらぎ

Lacey & Cole 1993, MN, 262, 627 , Bond et al 1991 ApJ. 379, 440
Peacock & Heavens 1990, MN 243, 133

2.3 Statistics of density fluctuations

PS factor 2 の "fudge"

- orign of angular momentum Hoyle Peebles 1969

tidal field / torque $F_+ \sim R^{-3}$ $|E| \approx \frac{GM^2}{R}$

spin parameter $\lambda = \frac{\tau_{dyn}}{\tau_{rot}} = \frac{\omega_{rot}}{\omega_{dyn}} \simeq \frac{V/R}{\sqrt{G\rho}} \simeq \frac{J/MR^2}{\sqrt{GM/R^3}} \simeq \frac{J|E|^{1/2}}{GM^{5/2}}$

$P(\lambda)d\lambda \to$ log normal See Fig

$\bar{\lambda} = 0.05$

$\sigma_\lambda = 0.5$

指向銀河 $\lambda \sim 0.05$ S ~ 0.5

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NO. 4

2.4. 散逸系の物理と銀河形成

hierarchical marging of helos

gravity alone → 任意の $\frac{M}{R}$ の halo ができる

実際は特徴的なスケール　Rees & Ostriker 1977, MN 179, 541

→ baryon physics 必要　Silk 1977, ApJ, 211, 638

$\frac{3}{2} n k T_{vir} = \frac{1}{2} \rho v^2 \quad v^2 = \frac{3GM}{5R} \quad \rightarrow T_{vir} = \frac{6GMm_H}{5R^2} = 10^6 \left(\frac{M}{10^{12} M_\odot}\right)\left(\frac{R}{100\,kpc}\right)^{-1} K$

hot gas ($\sim 10^6$K) ⟶ stellar system

radiative cooling

$t_{cool} = \frac{E}{\dot{E}} = \frac{\frac{3}{2} nkT}{n^2 \Lambda} \simeq 6.6 \times 10^5 yr \left(\frac{n}{cm^{-3}}\right)^{-1} \left(\frac{T}{10^6 K}\right) \left(\frac{\Lambda}{10^{-23} erg\, s^{-1} cm^3}\right)^{-1}$

$t_{ff} = \left(\frac{3\pi}{32 G\rho}\right)^{1/2} \simeq 5.2 \times 10^7 yr \left(\frac{n}{cm^{-3}}\right)^{-1/2}$

$t_H = 2/3 H_0 (1+z)^{3/2} = 6.5 \times 10^9 (1+z)^{-3/2} h^{-1} yr$

$(\Omega_0 = 1, \lambda_0 = 0)$

3. 楕円銀河の形成と進化

D Dec 7

↑

pop II, red, Giant Branch, Spheroid { Boxy ... brighter, flat core, slow rotation, older?
disky ... fainter, power-law cusp, fast rotation, younger?

various scaling relations { color-magnitude relation
　　↘ σ (速度分散)
Mg2 index − σ
Fundamental plane

3.1 Ages and metallicities of stellar populations

model vs obs $z = 0 \sim 1$

single burst

passive evolution

3.2 Scenarios of E formation

Tinsley 1980 Fund. Cosmic Phys. 5, 287

Matteuci. F, 1996 〃 , 17, 283

isolated, homogeneous, one-zone model　monolithic collapse

vs

marging ・spiral + spiral

・hierachical marging

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次回は来年 1/11, 18

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3.2.1

D Dec 25

3 Hierarchical marging Kauffmann et al 1993 MN
Cole et al 1994

4. 円盤銀河の形成と進化

disk component : exponential stellar disk + flat rotation curve
spheroidal " : metal-poor halo, globular cluster, bulge MACHOs

4.1 銀河円盤の形成

Silk & Norman 1981

D Jan 11

spin parameter $\lambda = |J||E|^{1/2}/GM^{5/2}$

$E = \frac{GM^2}{R}$

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